

eateggsAI-30M v0.1

Experimental Low-VRAM GPT-Style LLM Whitepaper

Abstract

eateggsAI-30M is an experimental GPT-style Large Language Model (LLM) created as a low-VRAM AI challenge using consumer-grade hardware. The project explores whether transformer-based language models can be trained efficiently on older GPUs with limited VRAM. The model was trained from scratch using PyTorch on a GTX 1060 4GB GPU while maintaining stable transformer training and meaningful text generation capabilities.

1. Introduction

Modern AI systems are commonly trained using expensive high-performance GPUs and large-scale infrastructure. This project investigates the feasibility of training a GPT-style transformer model on older consumer hardware with limited memory capacity.

eateggsAI-30M was designed as an educational and experimental research project focused on: Understanding transformer architectures deeply Building a custom LLM pipeline from scratch Exploring low-VRAM AI optimization techniques Training transformer models on affordable hardware

2. Hardware Configuration

Component	Specification
Laptop	Acer i5
GPU	NVIDIA GTX 1060
VRAM	4GB

3. Model Architecture

Setting	Value
Parameters	30,044,544
Transformer Layers	6
Attention Heads	8
Embedding Size	384
Context Length	256
Dropout	0.1

4. Dataset Pipeline

The training dataset was created using multiple domains in order to expose the model to diverse language structures and topics.

Dataset Sources: Wikipedia AG News SQuAD PG19 Books Web Text **Pipeline Steps:** Dataset downloading and streaming Data cleaning and filtering Deduplication GPT2 tokenization Fixed-length block creation PyTorch tensor

conversion

5. Training System

The model was trained using PyTorch with multiple optimization techniques for efficient low-VRAM training.

Training Features: Mixed Precision Training (AMP) Gradient Scaling AdamW Optimizer Cross Entropy Loss CUDA Acceleration Low-VRAM Optimizations

6. Experimental Results

Metric	Value
Training Time	~12 Hours
VRAM Usage	~2.7GB
Framework	PyTorch
Precision	Mixed Precision (AMP)
Final Loss	3.2 - 3.3

7. Conclusion

eateggsAI-30M demonstrates that transformer-based language models can still be trained effectively on older consumer hardware with careful optimization and efficient pipeline design.

This project highlights the accessibility of AI research and shows that meaningful experimentation with transformer architectures is possible without enterprise-level hardware.

8. Future Work

Improved text generation quality Larger model experiments Inference optimization Fine-tuning support Chat interface integration Advanced attention optimization methods

Project Links

GitHub: <https://github.com/eateggs0989/eateggsAI-30M>
Hugging Face: <https://huggingface.co/eateggs0989/eateggsAI-30M>